MANIPAL UNIVERSITY COLLEGE MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE

Histology of cornea, retina & optic nerve

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 # Only for instruction purpose..

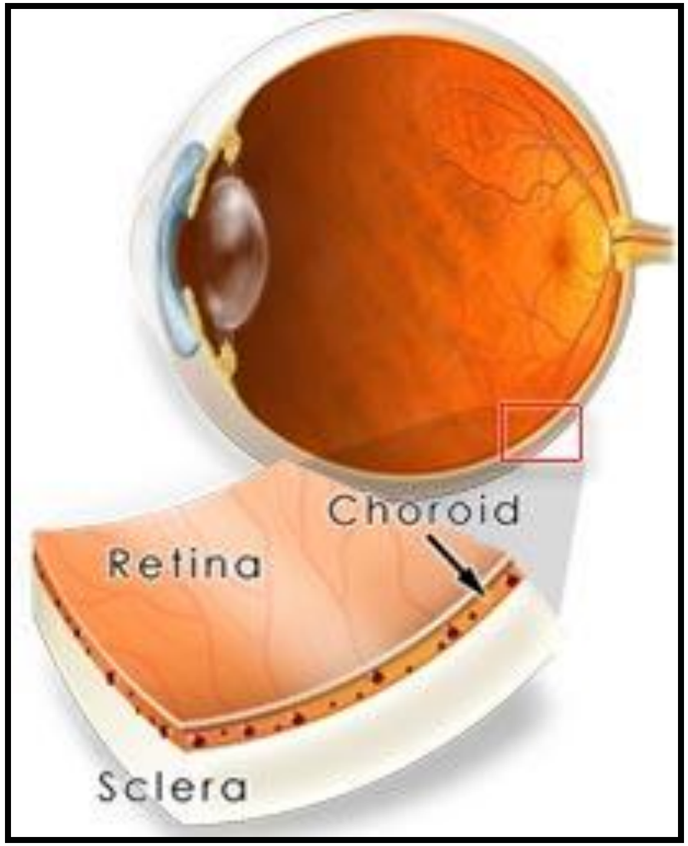
#inspire

CORNEA & RETINA

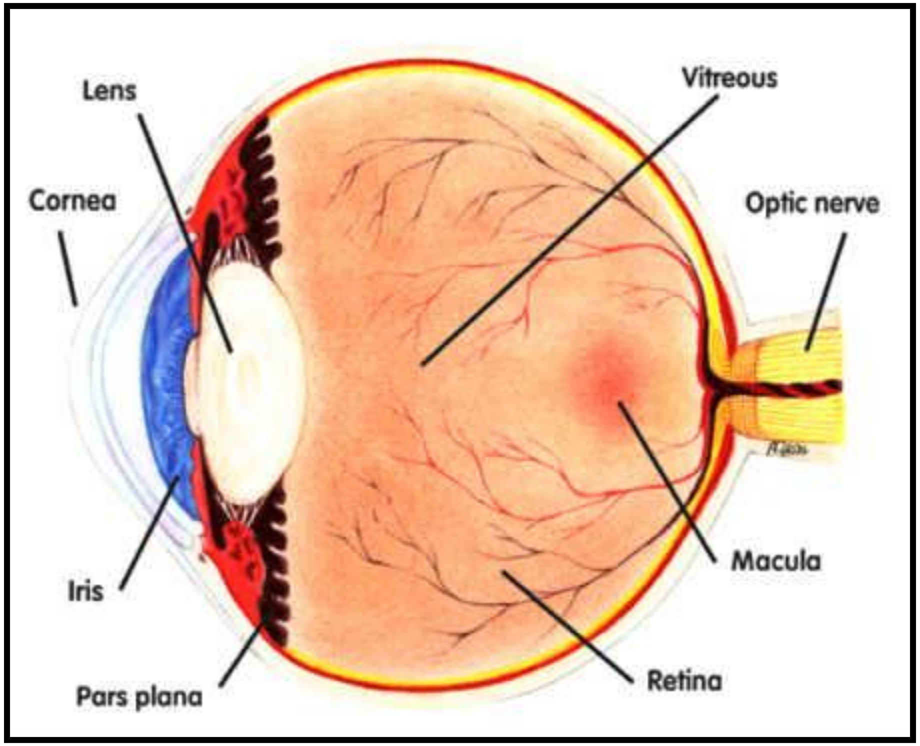
Learning objectives

Explain and illustrate the microscopic structure of cornea, retina and optic nerve

Coverings of eyeball



Parts of Eyeball



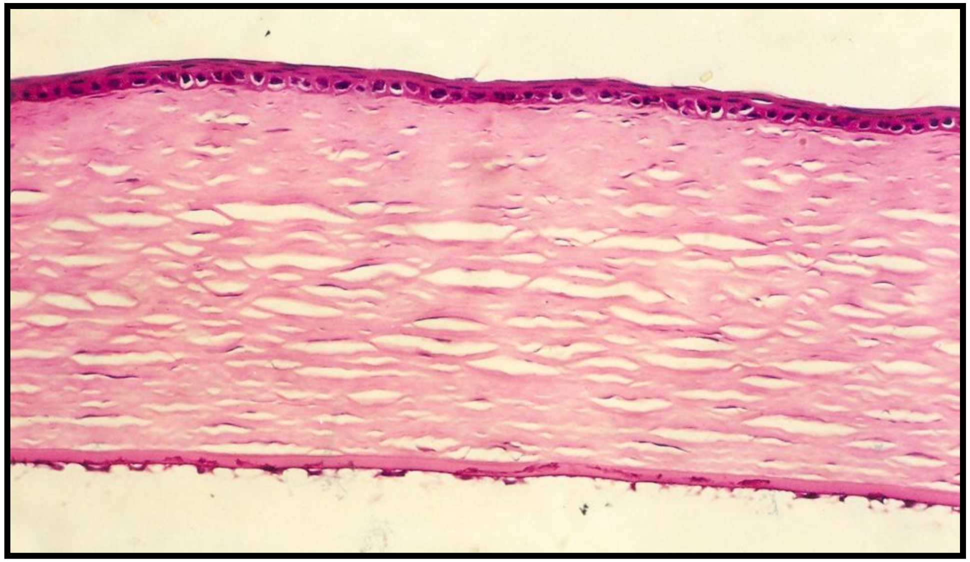
Cornea

- Five layers
- **Outermost layer-**

-Non-keratinized stratified squamous epithelium

- Deepest layer
- Middle layer
- Superficial layer

- **Great regularity**



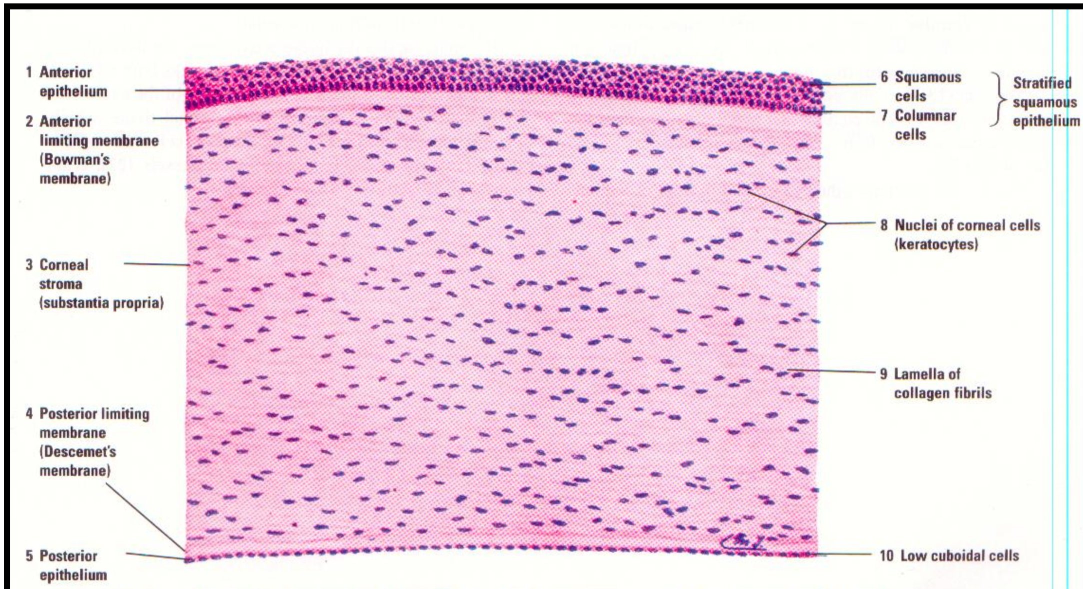


Fig. 20–3 **Cornea (transverse section).** Stain: hematoxylin-eosin. Medium magnification.

- Cells of superficial surface has microvilli or folds of plasma membrane
- They support the **glycocalyx** (a carbohydrate-rich coating) which allows the tear film to adhere to and spread evenly across the surface of the eye, keeping the cornea moist and optically clear
- Regeneration power - high

Anterior Limiting Lamina (Bowman's membrane

- Fine collagen fibrils embedded in matrix

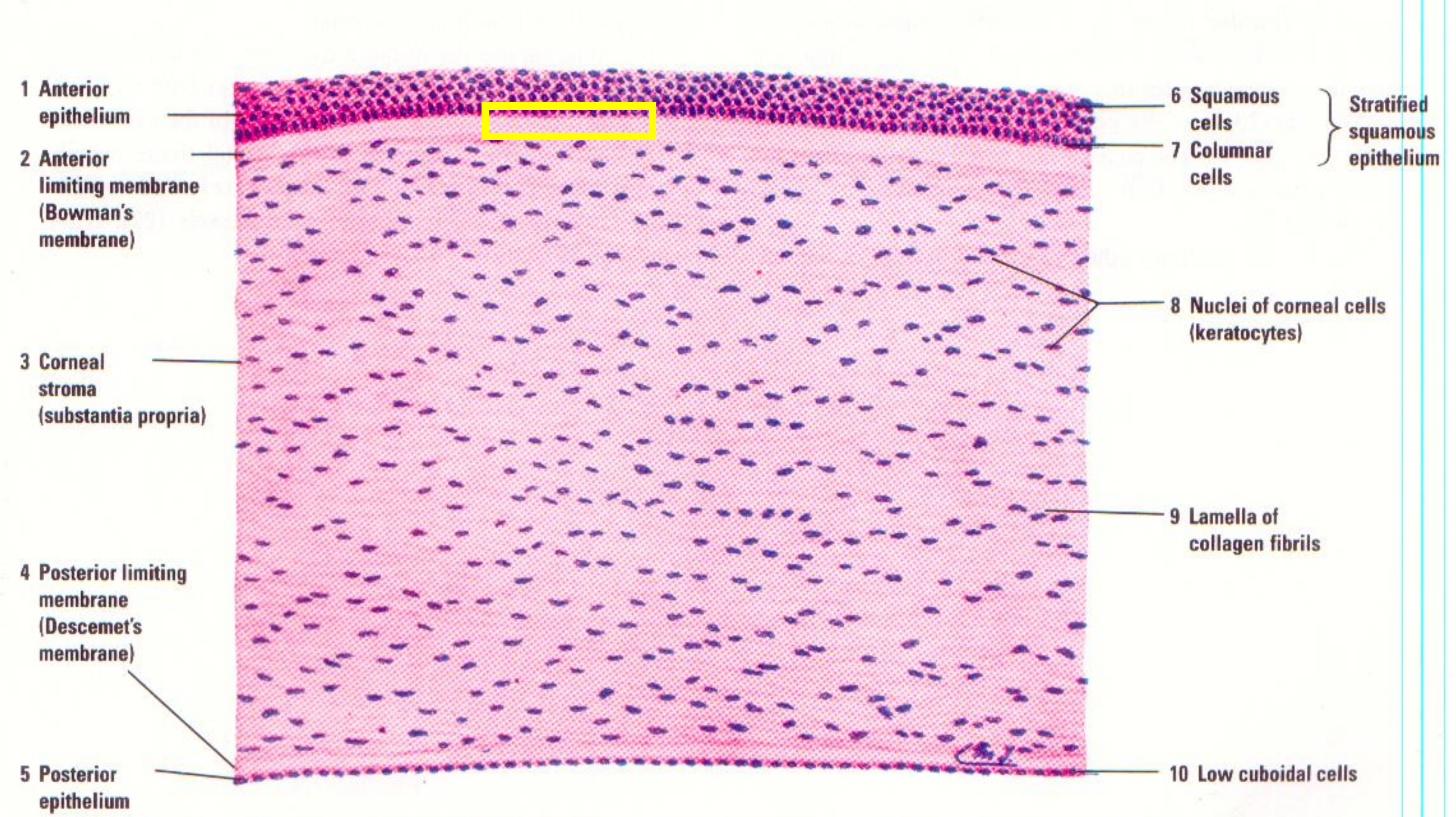


Fig. 20–3 **Cornea (transverse section).** Stain: hematoxylin-eosin. Medium magnification.

Corneal Stroma / Substantia Propria

- Collagen fibers –type-II
- Regularly arranged
- Glycosaminoglycans
- Same refractive index of ground substance and collagen fibers
- Fibroblasts & Keratocytes or Corneal corpuscles

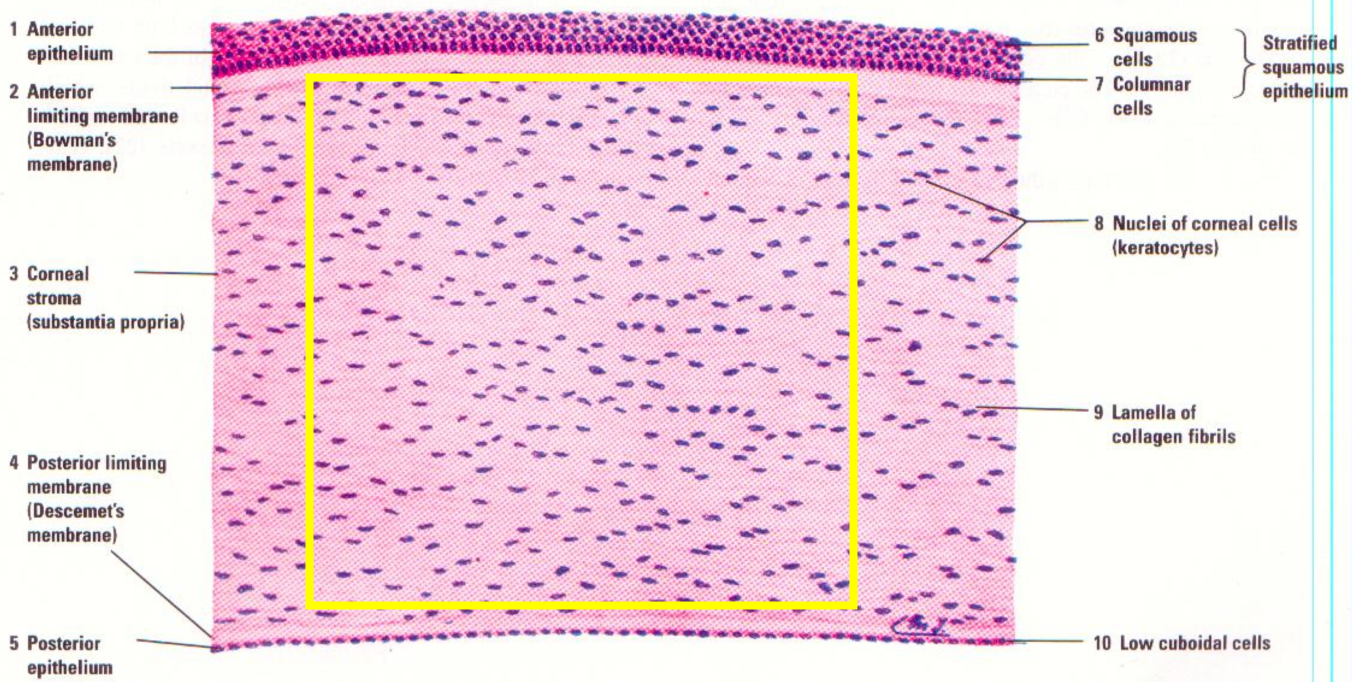


Fig. 20–3 **Cornea (transverse section).** Stain: hematoxylin-eosin. Medium magnification.

Descemet's Membrane / Posterior limiting lamina

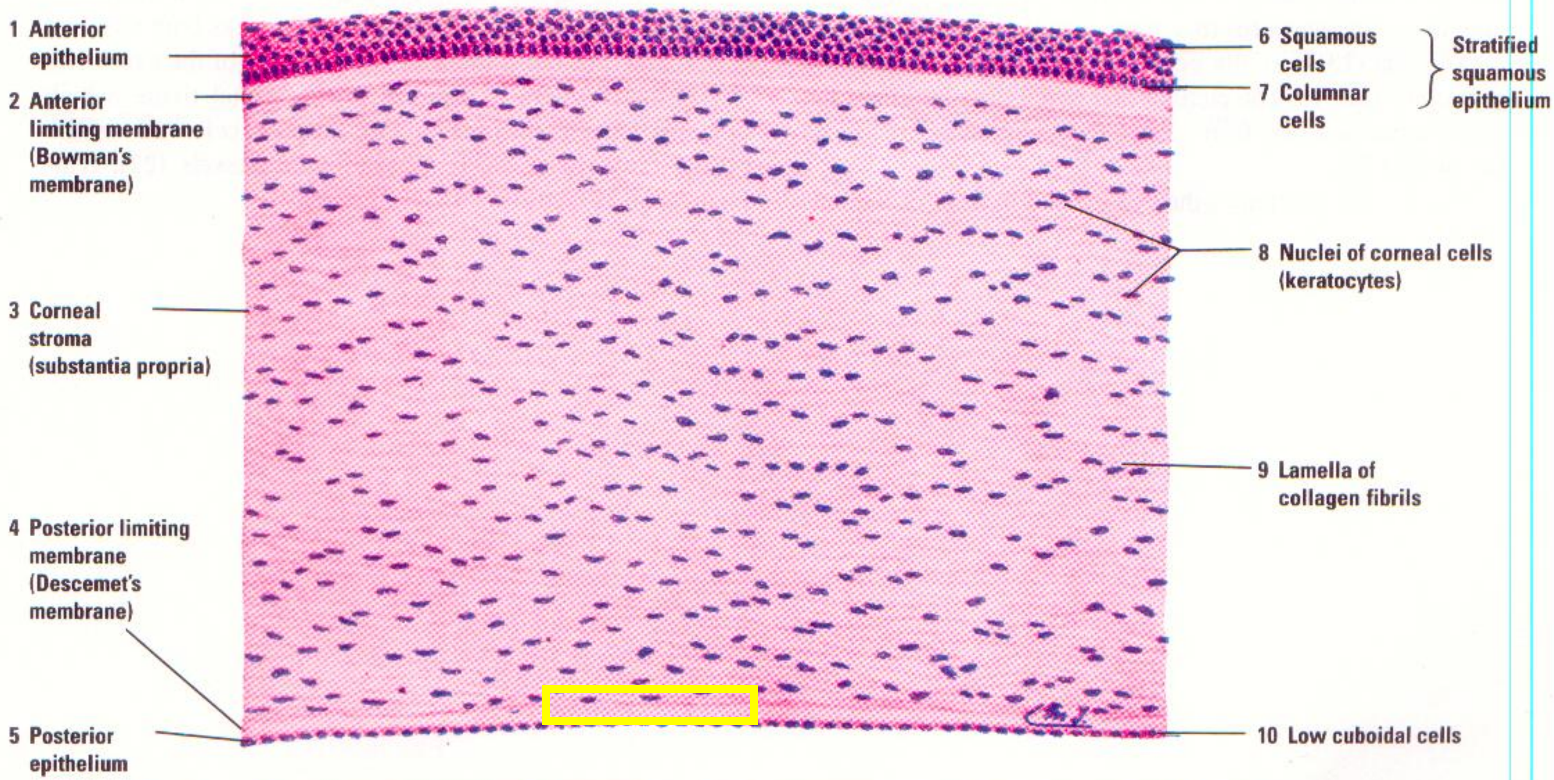


Fig. 20–3 **Cornea (transverse section).** Stain: hematoxylin-eosin. Medium magnification.

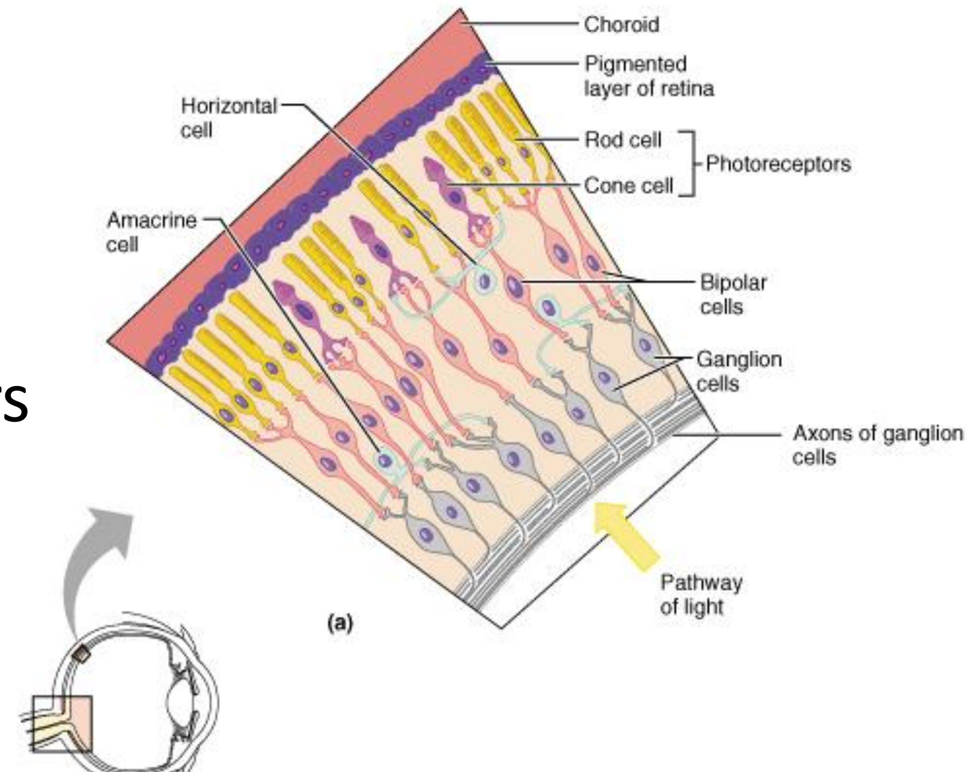
Endothelium

- Single layer of flattened cells
- Transport
- Numerous mitochondria
- Desmosomes
- Pump out excessive fluid
- Blood vessels and lymphatics are absent
- Rich nerve supply
- Nerve fibers are non-myelinated

Retina

Layers of retina

- Pigment cell layer
- Layer of rods and cones
- External nuclear layer
- External plexiform layer
- Internal nuclear layer
- Internal plexiform layer
- Layer of ganglion cells
- Layer of optic nerve fibers



Retina, Choroid, and Sclera

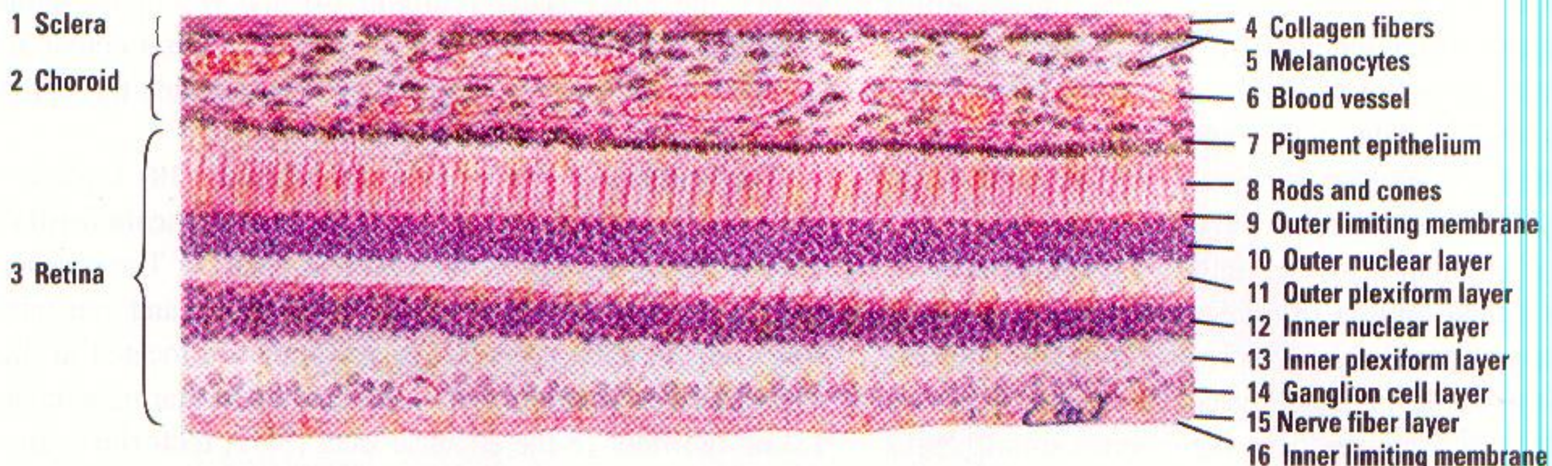
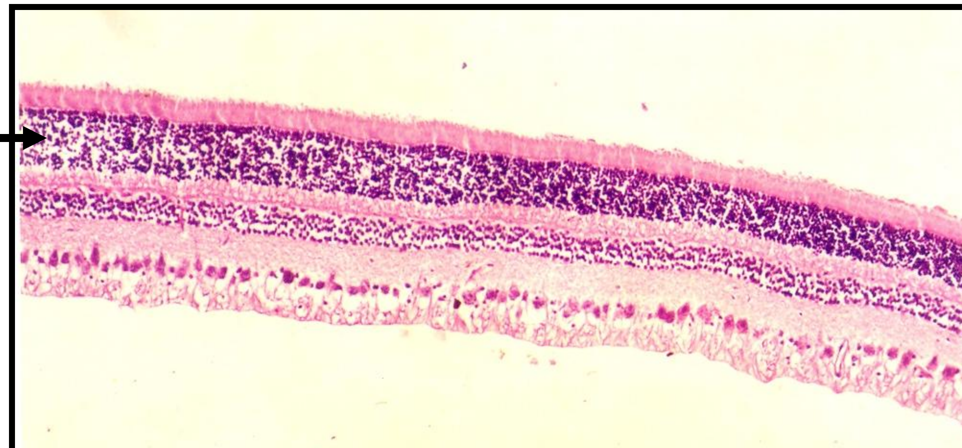


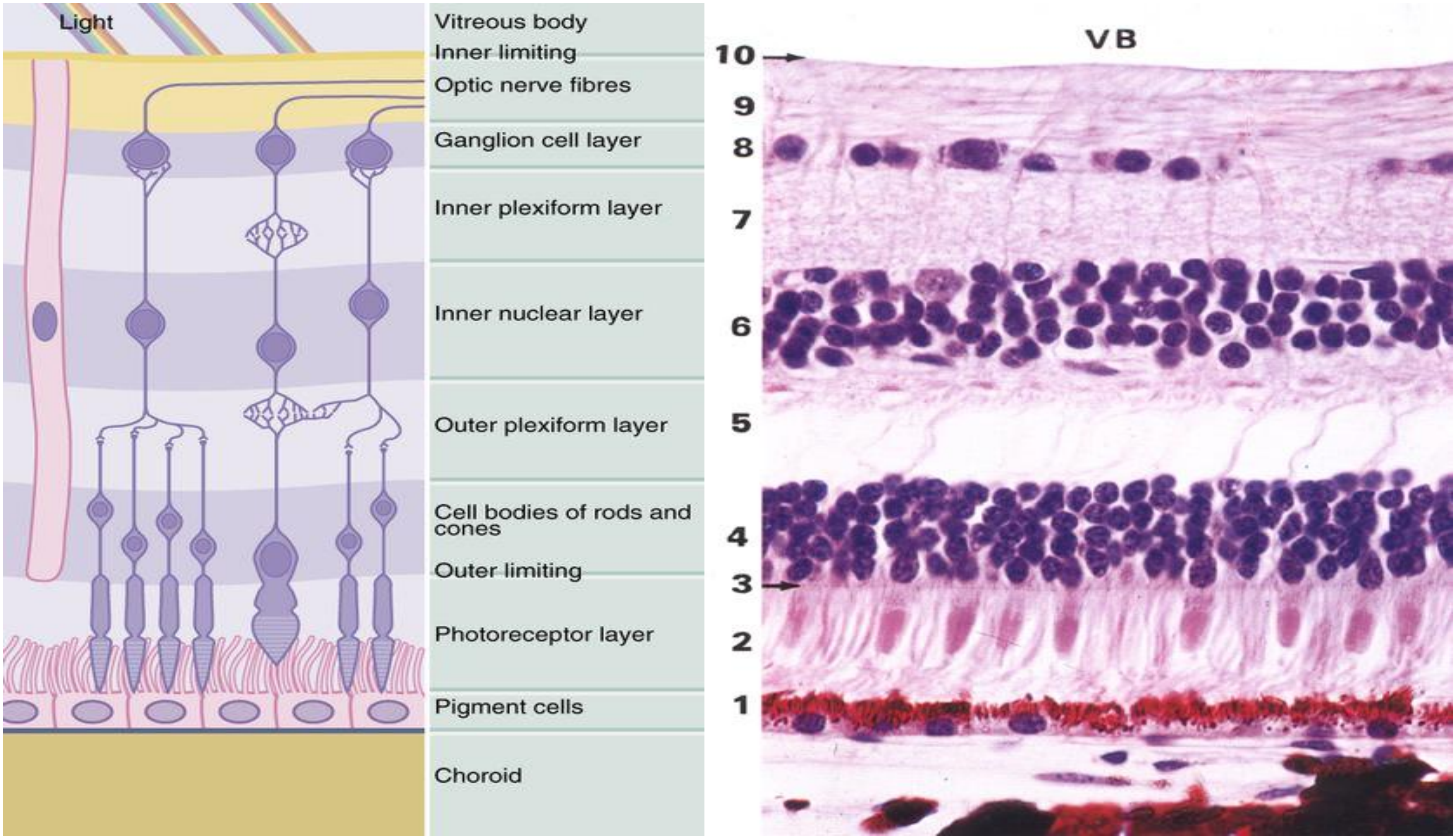
Fig. 20–5 **Retina, Choroid, and Sclera (panoramic view).** Stain: hematoxylin-eosin. Medium magnification.

<u>**Outer Nuclear Layer - Thick**</u>



Pigment cell layer

- Single layer – Pigment
- Has processes
- Absorption of excessive light
- Prevents back reflection
- Regular spacing
- Mechanical support
- Phagocytic role



Light
Vitreous body
Inner limiting
Optic nerve fibres
Ganglion cell layer
Inner plexiform layer
Inner nuclear layer
Outer plexiform layer
Cell bodies of rods and cones
Outer limiting
Photoreceptor layer
Pigment cells
Choroid
VB
10
9
8
7
6
5
4
3
2
1

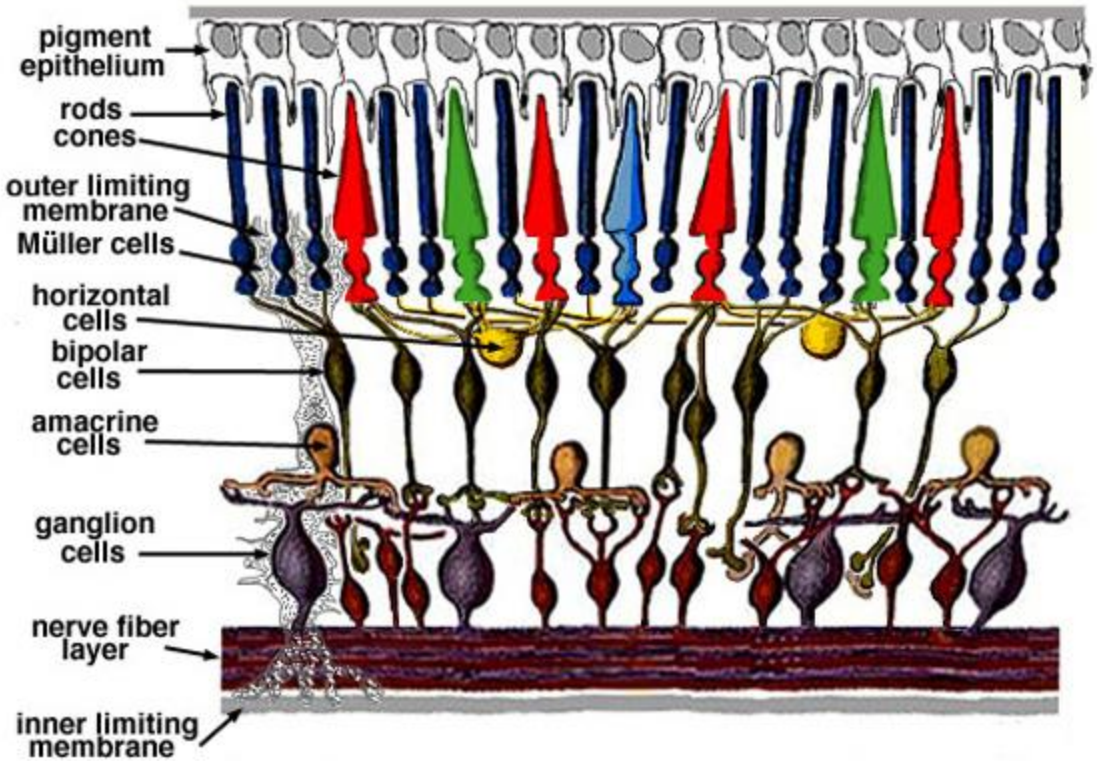
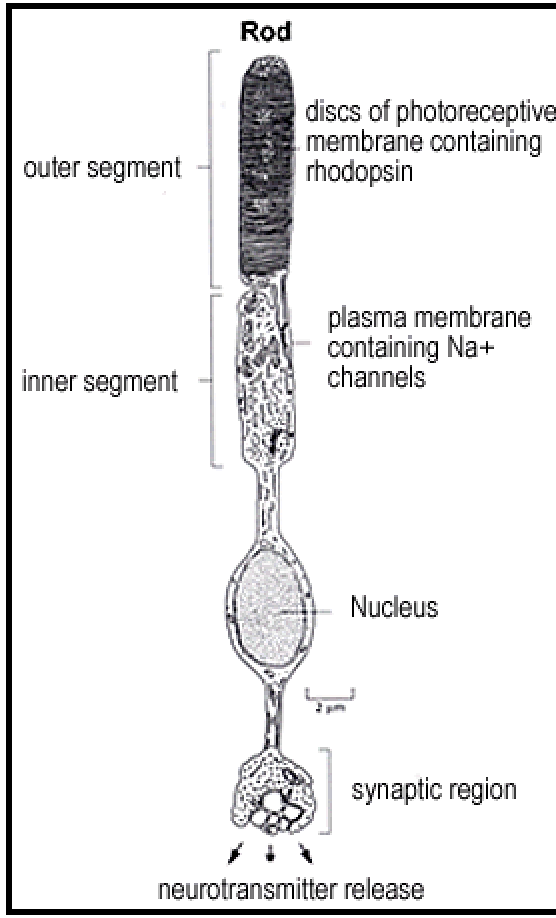


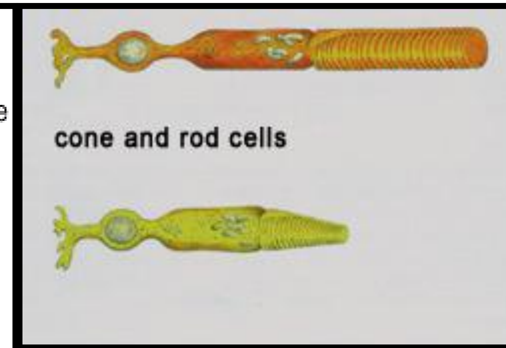
Fig. 2. Simple diagram of the organization of the retina.

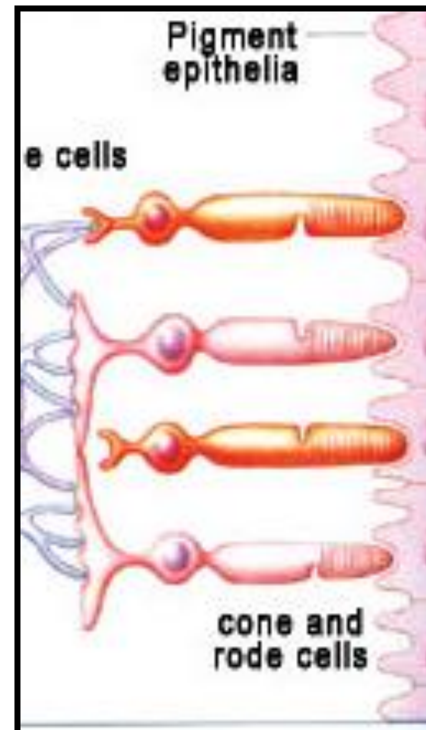
Rods

Cones

Tips – processes of pigment cells



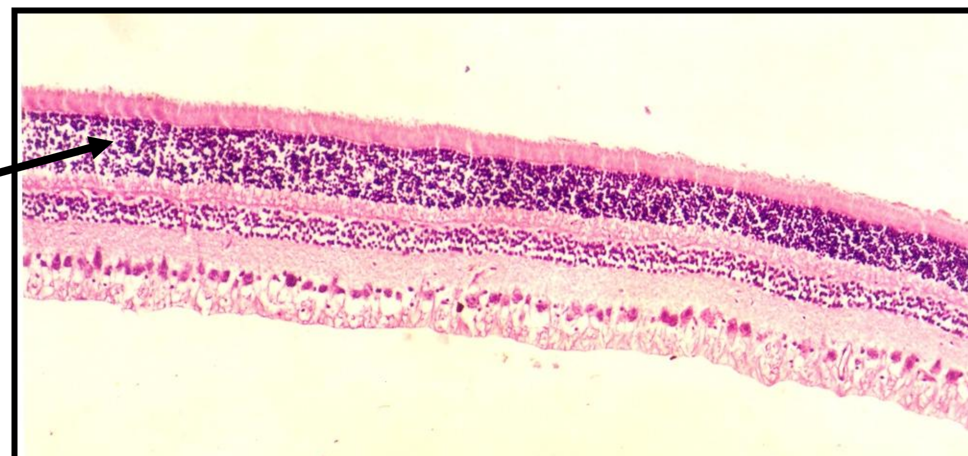




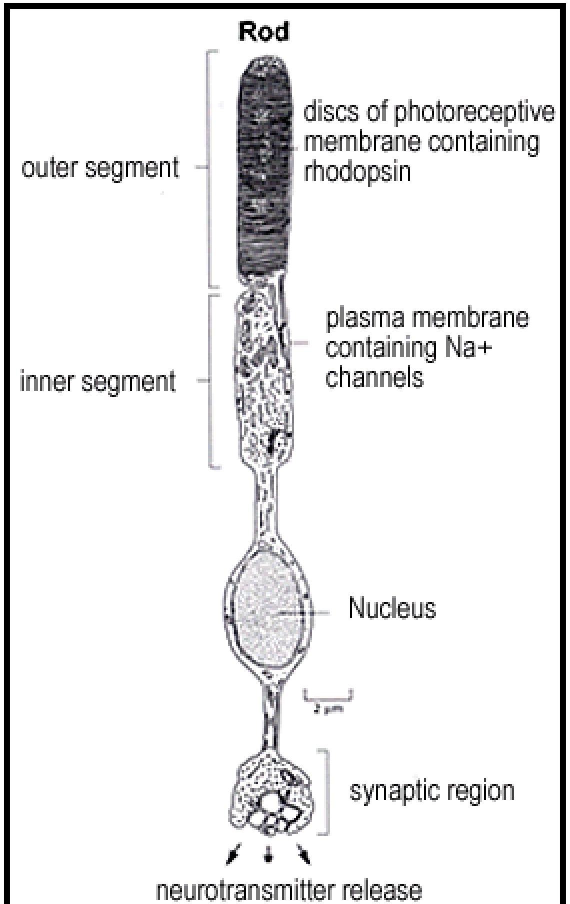
External Nuclear Layer

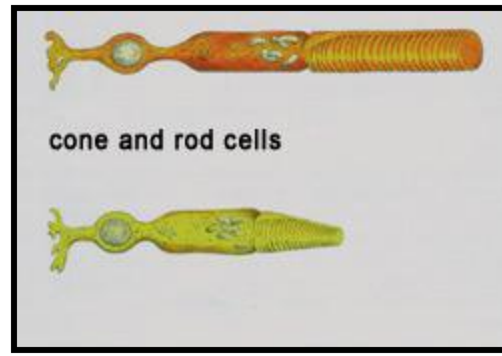
- Cell bodies of nuclei of Rod cells n Cone cells

Outer Nuclear Layer - Thick



Parts of Rod cells n Cone cells





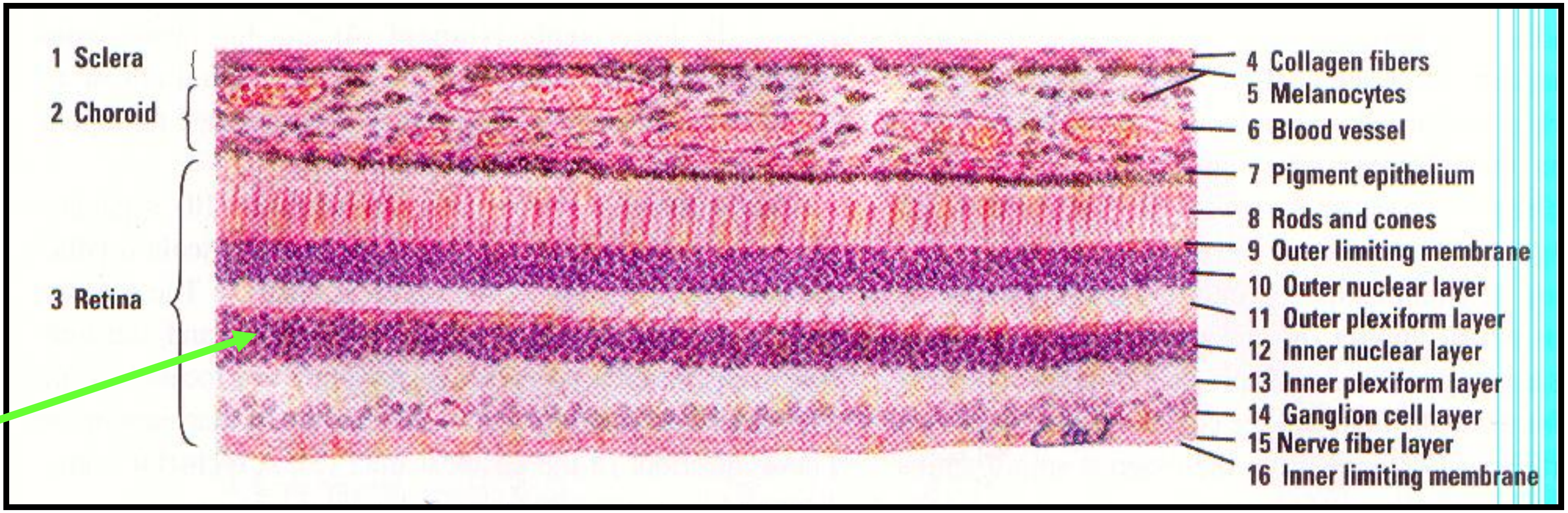
External Plexiform layer / Outer Synaptic zone

- Nerve fibres only
- Horizontal cells
- Axons of Rod cells n Cone cells synapses with dendrites of Bipolar neurons

Internal Nuclear Layer

- Cell Bodies of

1. **Bipolar neurons**
2. Horizontal cells
3. Amacrine cells



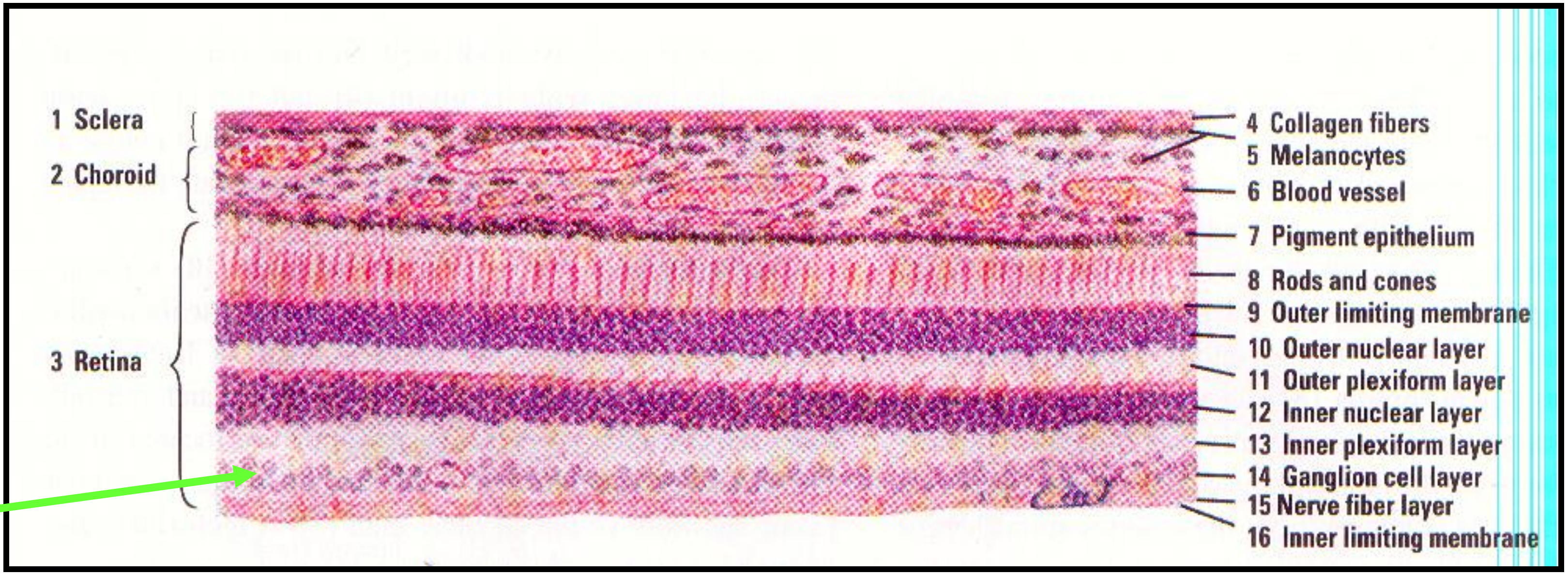
Internal Plexiform Layer / Inner Synaptic

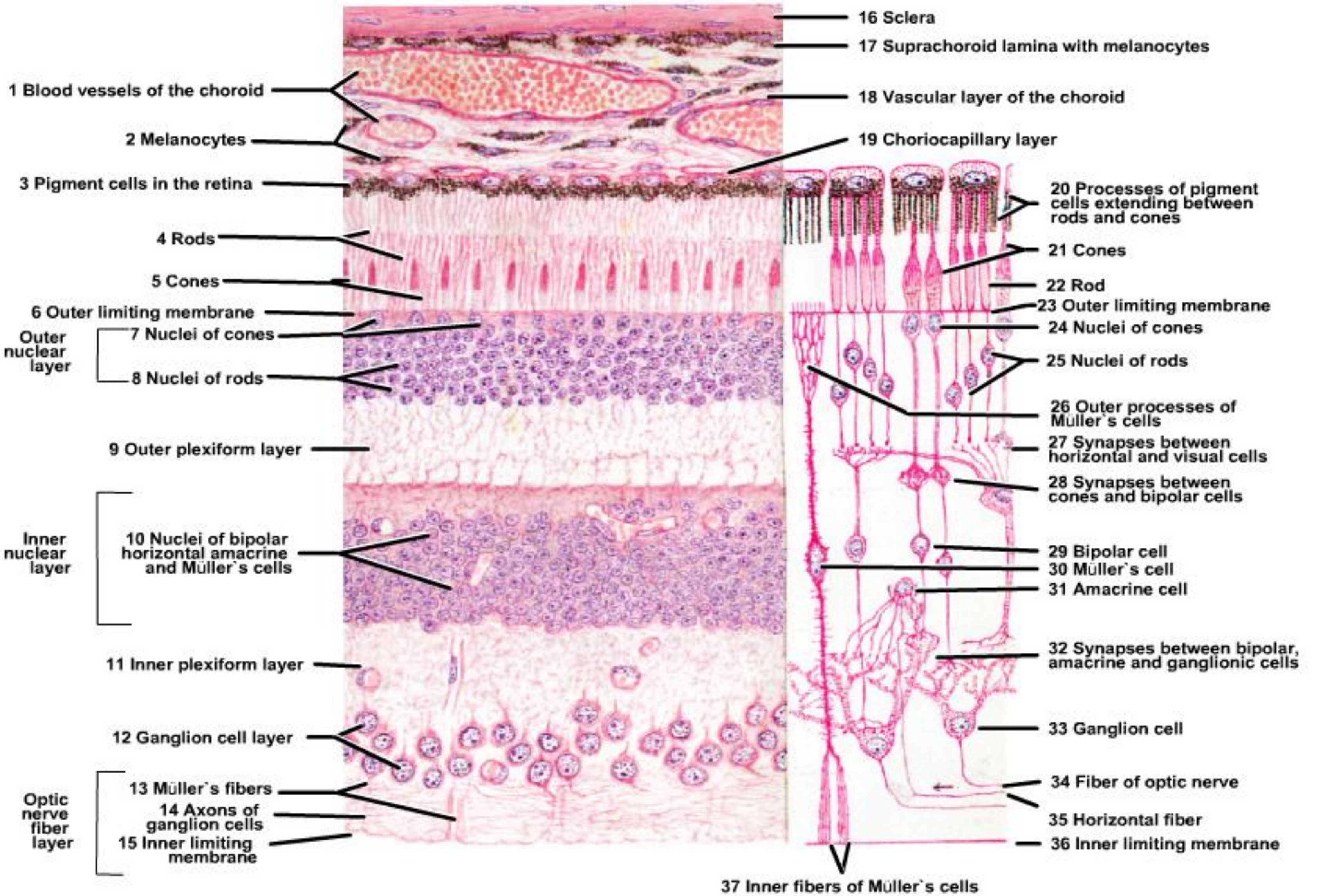
- Axons of Bipolar cells synapse with dendrites of ganglion cells

Fig. 20–5 **Retina, Choroid, and Sclera (panoramic view).** Stain: hematoxylin-eosin. Medium magnification.

Layer Of Ganglion Cells

- Cell bodies of Ganglion cells
- Axon – fibre of **Optic** nerve
- **2nd order neuron**




1 Blood vessels of the choroid
2 Melanocytes
3 Pigment cells in the retina
4 Rods
5 Cones
6 Outer limiting membrane
Outer nuclear layer
7 Nuclei of cones
8 Nuclei of rods
9 Outer plexiform layer
Inner nuclear layer
10 Nuclei of bipolar horizontal amacrine and Müller`s cells
11 Inner plexiform layer
12 Ganglion cell layer
Optic nerve fiber layer
13 Müller`s fibers
14 Axons of ganglion cells
15 Inner limiting membrane
16 Sclera
17 Suprachoroid lamina with melanocytes
18 Vascular layer of the choroid
19 Choriocapillary layer
20 Processes of pigment cells extending between rods and cones
21 Cones
22 Rod
23 Outer limiting membrane
24 Nuclei of cones
25 Nuclei of rods
26 Outer processes of Müller`s cells
27 Synapses between horizontal and visual cells
28 Synapses between cones and bipolar cells
29 Bipolar cell
30 Müller`s cell
31 Amacrine cell
32 Synapses between bipolar, amacrine and ganglionic cells
33 Ganglion cell
34 Fiber of optic nerve
35 Horizontal fiber
36 Inner limiting membrane
37 Inner fibers of Müller`s cells

Optic nerve

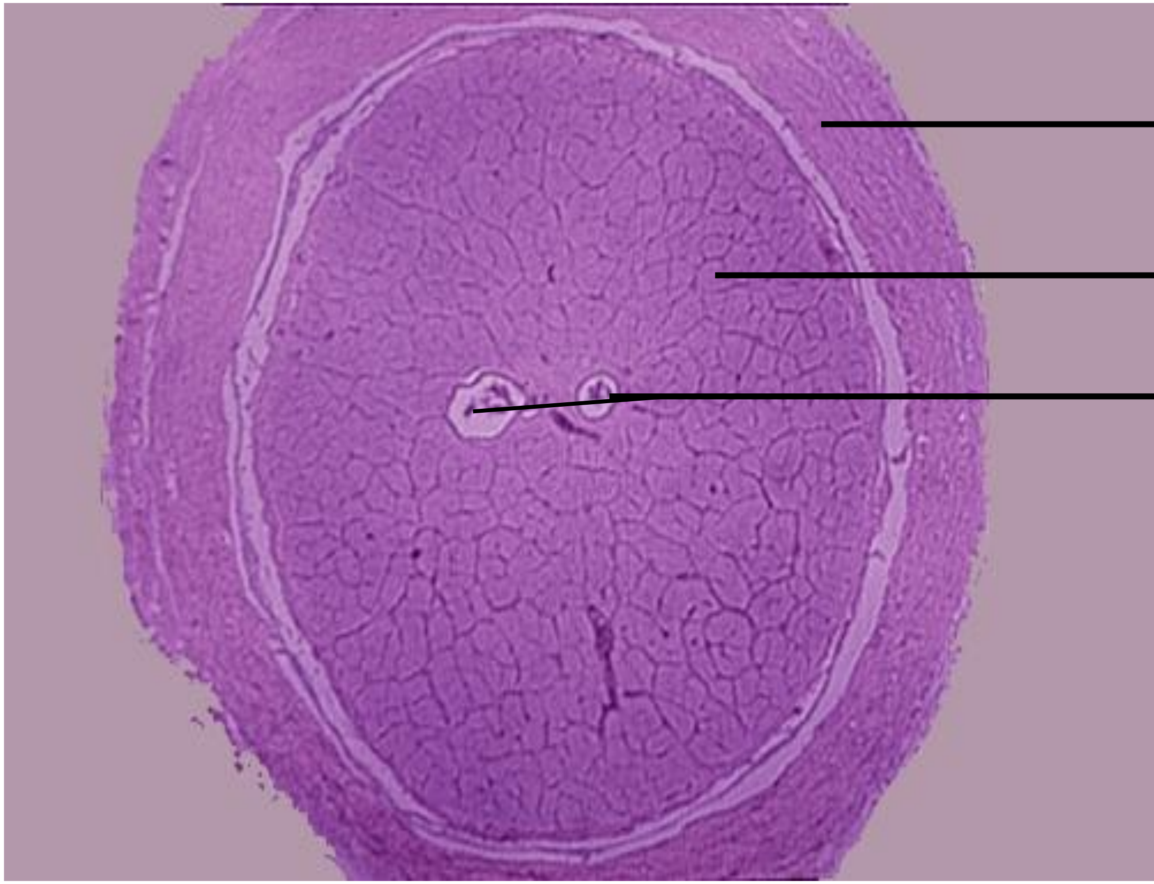
Core Histological Components

•**Axons:** Unmyelinated within the eye, these axons acquire a myelin sheath as they pass through the *lamina cribrosa* and travel to the brain.

•**Glial Cells:** Astrocytes provide structural and metabolic support, while oligodendrocytes are responsible for producing the myelin sheath.

•**Connective Tissue Septa:** Delicate extensions of the pia mater project inward to divide the nerve fibers into distinct bundles or fascicles.

•**Central Retinal Vessels:** The central retinal artery and vein run through the center of the nerve to supply blood to the optic structures.


Meningeal layers with spaces
Neuronal tissue
Artery and vein

Thank you